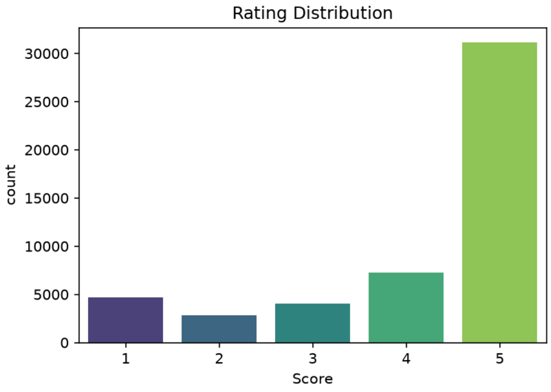
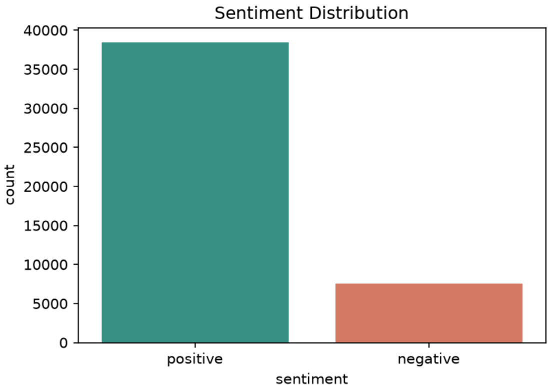
Customer Review Intelligence Report

Analysis of 49,998 Amazon fine food reviews.

1. Overview

Metric	Value
Total reviews	49,998
Positive (4-5 stars)	38,418
Negative (1-2 stars)	7,533
Neutral (3 stars)	4,047
Average rating	4.146

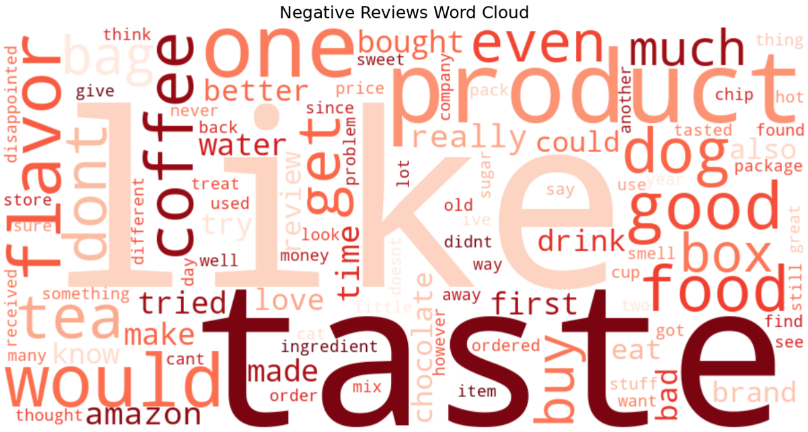




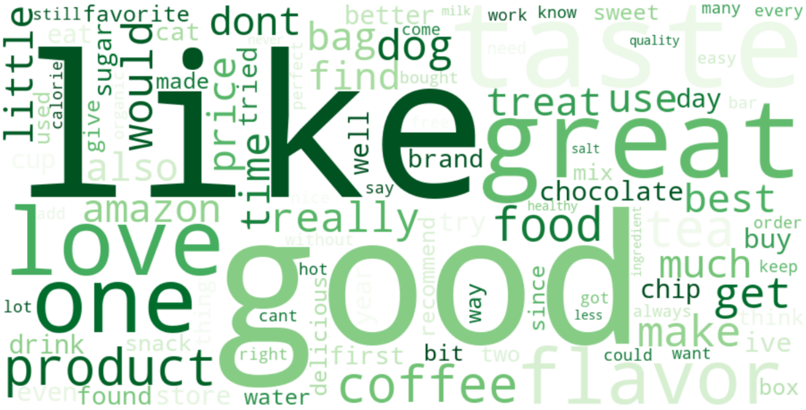
2. What customers talk about

Words most specific to negative reviews: refund, worst, expired, disappointing, fda, threw, disappointment, trash, return, awful, menadione, nespresso, yuck, horrible, disgusting

Words most specific to positive reviews: peppercorn, hooked, pleasantly, satisfies, tastiest, complement, strainer, pickmeup, magnesium, downside, calming, wonderfully, beautifully, yummy, watcher



Positive Reviews Word Cloud



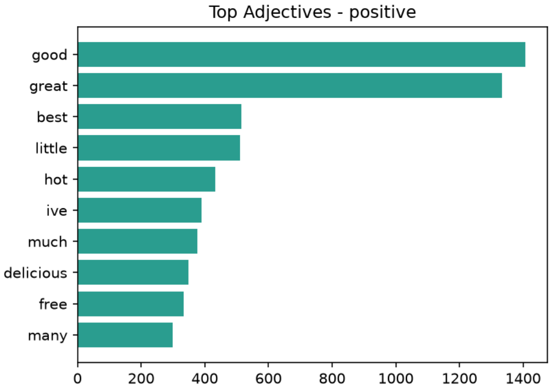
3. Topics

Complaint themes (LDA)

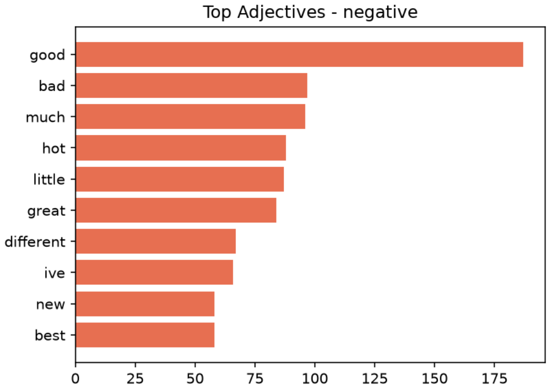
Topic 1: like, taste, chocolate, chip, bar, product, would, good
Topic 2: like, one, sugar, product, would, taste, ingredient, get
Topic 3: taste, like, coffee, tea, flavor, cup, drink, good
Topic 4: product, amazon, box, bag, one, item, order, price
Topic 5: dog, food, product, treat, cat, made, chicken, one

Praise themes (LDA)

Topic 1: coffee, tea, cup, flavor, like, taste, good, one
Topic 2: treat, dog, like, drink, sugar, taste, juice, flavor
Topic 3: bag, great, product, good, amazon, price, one, like
Topic 4: food, love, dog, one, like, cat, good, use
Topic 5: taste, like, great, good, love, product, chocolate, mix



Top Adjectives - positive

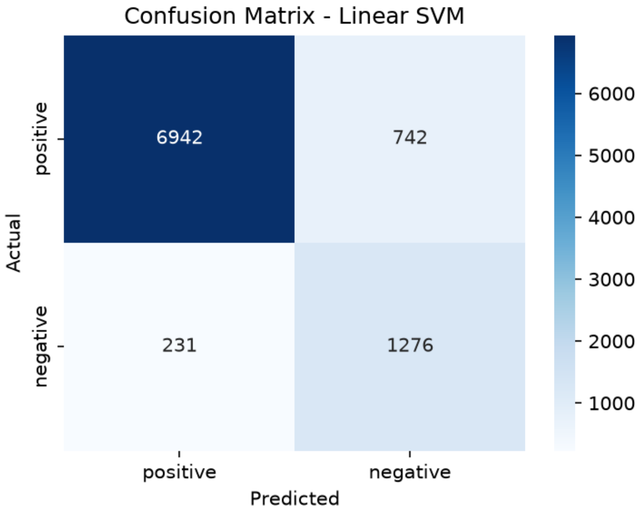


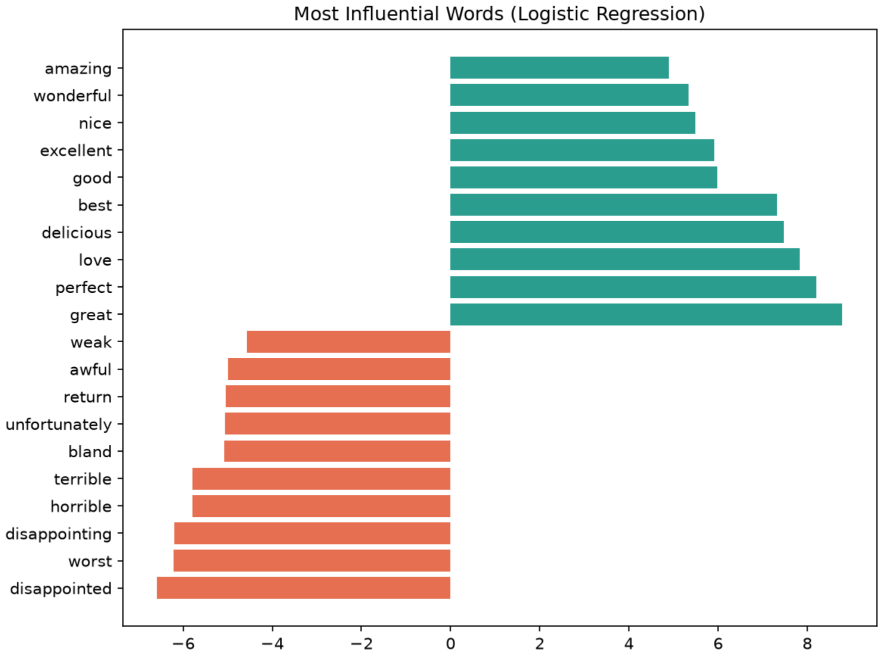
Top Adjectives - negative

4. Sentiment Classification

Best model: Linear SVM (selected on macro F1, because the data is about 84% positive and plain accuracy hides weak negative-class performance).

Model	Accuracy	Macro F1	Negative F1
Naive Bayes	0.8833	0.7061	0.4779
Logistic Regression	0.8891	0.8257	0.7206
Linear SVM	0.8941	0.8292	0.724





Limitations

Labels come from star ratings, not human annotation. Neutral (3 star) reviews were excluded from training, so the model tends to treat neutral text as negative.